

Chapter 4

German *werden*-future: A case of failed grammaticalization?

Elena Smirnova

Université de Neuchâtel

In present-day German, the construction *werden* + infinitive is usually considered as an analytic future tense form, and the diachronic development of this construction is described as a typical case of grammaticalization. However, the synchronic status of this construction is still a matter of discussion. On the one hand, the function of this construction is not restricted to expressing futurity, but encompasses modal and evidential functions. On the other hand, when it comes to marking futurity, *werden* + infinitive is not obligatory. Although the relatively high grammaticalization status of the construction seems uncontested, especially when compared with the less grammaticalized copula verb *werden* from which it originated, this paper will argue that the grammaticalization process has not come to full completion and was halted at some point in time. What happened then is that the use of the futurate *werden* + infinitive was gradually restricted to a rather small range of particular contexts of use, i.e. a functional niche, where it unambiguously serves as a future tense marker.

1 Introduction

It is rather uncontroversial that present-day German does not have a default future construction in the sense that (i) there exists a verbal periphrasis with a dedicated future auxiliary and (ii) this periphrasis is highly grammaticalized and (iii) it is obligatory in all contexts with future time reference. Though most grammar books of present-day German present *werden* + infinitive as a future tense construction (see 1) along with other analytic tenses such as perfect or pluperfect, this construction in fact only marginally functions as an analytical future tense.



- (1) *Er wird im Urlaub sein.*
he werden.3SG.PRES on vacation be.INF
'He WIRD be on vacation.'
a. He will be on vacation. = future time reference
b. He might be on vacation. = epistemic assessment
c. He is bound to be on vacation. = evidential assessment

Without going into much detail on the longstanding debate regarding semantic and functional characteristics of *werden* + infinitive in present-day German, it is fair to say that it is not an easy task to determine the function of this construction. In the literature, there are three main accounts that may be briefly summarized as follows:

One branch of researchers, the so-called "temporalists" (see e.g. Matzel & Ulvestad 1982; Thieroff 1992) consider *werden* to be primarily a future tense auxiliary (see the paraphrase in 1a) and derive other readings from this future semantics.

The so-called "modalists" (see esp. Vater 1975) on the other hand, treat *werden* essentially as a modal verb, namely as an epistemic modal verb, see the interpretation given in (1b); and other readings are then derived from the modal semantics of this verb.

A third option is to see *werden* as primarily evidential in meaning (see esp. Diewald & Smirnova 2010) namely an inferential evidential verb, as in (1c).

The present paper does not attempt to answer the question of what counts as the central or core meaning of the construction *werden* + infinitive. Instead, it focuses on the temporal reading of this construction and argues that this reading is restricted to a very marginal set of contexts which constitute a functional niche of this construction. The goal of this paper is first, to show that the grammaticalization of *werden* + infinitive has not been proceeding further since late Early New High German (ENHG), and second, to explore the functional niche it occupies in present-day German.

The paper is structured as follows. In Section 2, I will briefly recall some known facts about the diachrony of several future tense candidates in the history of German and motivate my research questions. In Section 3, I will report the results of a diachronic corpus study that concentrates on some particular contexts of use, namely subordinate temporal and conditional clauses. It is in these contexts that the hampered grammaticalization of the future marking manifests itself prominently. In Section 4, I will focus on synchronic corpus data in zooming in on another context of use, namely the combination of *werden* and the infinitive verb

sein ‘to be’, in order to pinpoint the characteristics of the functional niche the construction occupies in present-day German. The paper will conclude with some thoughts on the reasons why *werden* has not reached an advanced degree of grammaticalization in German.

2 Future constructions in the diachrony of German

In present-day German, *werden* + infinitive is used to mark posteriority, that is, to locate the described event in the future relative to the speech time, as for example in (2).

- (2) *Bereits am 1. April wird das Strandbad Müggelsee seine*
 already on 1st April *werden*.3SG.PRES the Lido Müggelsee its
Pforten öffnen.
 gate.PL open.INF
 ‘The Müggelsee Lido will open as early as April 1st.’
 [Berliner Tageblatt 1931, DWDS-Kernkorpus]

From the diachronic perspective, it is well known that *werden* + infinitive was not the only construction on the way to becoming a future marker in the history of German. Prior and partly in parallel to *werden* + infinitive, especially in Middle High German (MHG), the combination of *werden* and present participle was used to mark future time reference (cf. 3; see Hartmann 2021; Ihden 2023).

- (3) *Salic fit ir die nu weinint. wan ir*
 blessed be.2.PL.PRES you who now weep.2PL.PRES. when you
werdint lachinde.
werden.2PL.PRES laugh.PRES.PTCPL.PL
 ‘Blessed are you who weep now, for you shall laugh.’
 [REM¹, 12th c., from Hartmann (2021: 367)]

As regards *werden* + present participle, it became obsolete in the course of the ENHG period, as has been clearly shown by Stefan Hartmann in his diachronic corpus study, cf.:

Während im REM *werden* + Partizip die deutlich frequentere Konstruktion ist und beide Konstruktionen vom 12. zum 13. Jahrhundert einen Anstieg

¹REM stands for *Referenzkorpus Mittelhochdeutsch*, a corpus of Middle High German data (see <https://www.linguistics.rub.de/rem/> for more details).

erfahren, sinkt die Gebrauchsfrequenz der *werden* + Partizip-Konstruktion vom 13. zum 14. Jahrhundert deutlich. [...] *werden* + Infinitiv erlebt einen geradezu explosionsartigen Anstieg, während *werden* + Partizip im eher niedrigen Frequenzbereich verharret. (Hartmann 2021: 377)²

But also the modal verbs such as *sollen* ‘shall’, *wollen* ‘want, will’, and *müssen* ‘must’ were all used with future time reference in the history of German (cf. 4; see Diewald & Habermann 2005 for further details).

- (4) *Ach Gott hilff* / *Ach Got hoere* / *Ach Got* / *wenn*
 oh God help.IMP.SG / oh God hear.IMP.SG / oh God / when
will *es ein end werden?*
 will.3SG.PRES it a end werden.INF
 ‘Oh God help / Oh God hear / Oh God / when will it end?’
 [Dietrich, 16th c., Diewald & Habermann (2005: 232)]

As for the modal verbs, they too did not develop future marking as their central semantics, and did not proceed further on the grammaticalization path towards future constructions in German (cf. Diewald & Habermann 2005: 232): “Die Entwicklung von *werden* zum Futurmarker hat ganz offensichtlich den Grammatikalisierungsprozess der Modalverben zu Futuren gebremst”.³

Werden + infinitive is hence a clear winner in this diachronic competition with modal verbs and with the construction *werden* + present participle, as only this construction successfully developed into a future tense marker. However, the story of *werden* + infinitive appears to be less successful as it might seem at first glance.

For present-day German, it has been repeatedly observed that this construction is generally very infrequent. To refer to future time, German speakers clearly prefer present tense forms (cf. Brons-Albert 1982; Hilpert 2008). For example, Brons-Albert (1982) found that only 30% of all futurate contexts contain *werden* + infinitive. The unmarked expression of future time today is not *werden* + infinitive, but the (furate) present tense.

²Translation: While *werden* + participle is the considerably more frequent construction, and both constructions experience an increase from the 12th to the 13th century, the usage frequency of the *werden* + participle construction decreases considerably from the 13th to the 14th century. [...] *werden* + Infinitive experiences an enormous increase, while *werden* + participle remains in a rather low frequency area.

³Translation: The development of *werden* to a future marker has quite obviously curbed the grammaticalization process of modal verbs to futures.

Thus, it seems that no significant semantic and formal changes have occurred since *werden* + infinitive prevailed over the other constructions in the late Middle High German period, and, most importantly, the frequency of use has remained very low or even decreased since then. In the remainder of this paper, I will argue that this construction has not progressed further on the way towards becoming a highly grammaticalized construction. Instead, it has even regressed, and is now limited to some specialized niches.

In the following, I will focus on two local contexts of use of this construction, which I will examine further in order to substantiate my claim:

1. From the diachronic perspective, I will look at conditional and temporal subordinate clauses, i.e. the contexts in which *werden* + infinitive can be unambiguously interpreted as conveying future time reference. Starting from the assumption that the appearance of *werden* + infinitive in conditional clauses represents an advanced step in the grammaticalization process towards a future tense marker, a detailed exploration into these contexts will give us valuable insights into the diachronic development of this construction.
2. From the synchronic perspective, the contexts with the infinitive verb *sein* 'be' will be dealt with in a more detailed quantitative and qualitative analysis of corpus data from present-day German. As previously shown in studies by Hilpert (2008) and Hartmann (2021), and as will be confirmed by my own diachronic analysis in Section 3, this verb is the most frequent infinitive verb that combines with *werden*. Given the fact that *sein* itself is a highly polyfunctional verb in present-day German, the mere co-occurrence of the two does not provide us with any substantial information about the functional and/or semantic range of *werden* + *sein*. The in-depth analysis of these contexts presented in Section 4 will identify the relevant characteristics of the semantic niche *werden* + infinitive occupies in present-day German.

3 *Werden* + infinitive in conditional and temporal clauses

In a comprehensive diachronic corpus study of *würde* + infinitive in MHG and ENHG (Smirnova 2006), I also looked at the indicative form of this construction, i.e. *werden* + infinitive. One of the central insights of this study was that, in its diachronic development, *werden* + infinitive passed through an intermediate stage

of semantic change where it clearly was used to mark a textually salient sequential relation. *Werden* + infinitive acquired this semantics in the MHG period via the reinterpretation of the original *aktionsart* ingressive meaning (see Smirnova 2006: 259–267 for further details).

In this respect, the development of *werden* + infinitive towards a future tense construction follows a rather unusual trajectory. Starting as a copula verb with aspectual change-of-state semantics (i.e. ingressive *aktionsart*), it developed into a marker of textual sequence relation before turning into a temporal marker of posteriority. These semantic shifts proceeded incrementally and gradually along a series of syntactic contexts, and they show a particular directionality: (i) from main clauses preceded by subordinate clauses to (ii) independent clauses and (iii) finally to subordinate clauses.

In main clauses preceded by subordinate clauses, i.e. in stage (i) of the development, the proposition introduced by *werden* + infinitive is marked as following from some important pieces of information given in the text, as shown for example in (5) and (6).

- (5) *Denn wenn solch stück der Königin Auskopmt zu den weiberen hin, So*
then when such piece the queen come.out.3.SG to the women to so
werden *sie verachten gar All ir menner ganz offenbar.*
werden.3.PL they despise.INF even all their men quite obviously
‘For if such a piece of the queen’s head is sent out to the women, they
will despise all of their men quite obviously.’

[ENHGCORPUS, Smirnova (2006: 262)]

- (6) *Wer nun Tugend / gute Sitten vnd Kuensten liebhat/ wird*
who now virtue / good morals and skills like.3.SG/ werden.3.SG
wissen was er tun sollte.
know.INF what he do.INF shall.3.SG
‘Now he who loves virtue / good morals and arts / will know what he
should do.’

[ENHGCORPUS, Smirnova (2006: 263)]

In (5), *werden* + infinitive occurs in the apodosis of a conditional sentence and introduces a conditional consequence. In (6), on the other hand, *werden* + infinitive introduces an effect of a condition that appears in the form of a free relative clause. Additionally, in both examples (5) and (6), this construction can be interpreted as referring to future time. What is important is that in these contexts, the predicates do not refer to a specific future event, but they refer generically to certain types of events.

In the course of ENHG, the temporal future reading of the construction *werden* + infinitive gets gradually emancipated from this kind of textual support and expands into new contexts of use. This is especially the case for uses of the construction in independent clauses, see (7). In these contexts, the marking of textual sequence relation is no longer the function of this construction.

- (7) *Von dem andern geschriben ist: Dann **wierst** du **nemmen***
of the other write.PAST.PART be.3SG: then werden2.SG you take.INF
das oppher der gerechtichait.
the sacrifice the justice
‘Of the other it is written: Then you will take the sacrifice of justice.’
[ENHGCORPUS DR, Smirnova (2006: 261)]

The next and most advanced step in the semantic development of future tense marking is reached when the construction unambiguously bears future time reference and does not bear any textual relation to the preceding text. This reading is most pronounced in the subordinate temporal and conditional clauses, as shown in (8)–(10).

- (8) *Dem Lande wird wiederum wohl seyn **wenn** Lycogenes die*
the country werden.3.SG again happy be.INF when Lycogenes the
*Krone tragen **wird***
crown wear.INF werden.3.SG
‘The country will again be happy when/if Lycogenes will wear the crown.’
[ENHGCORPUS 147, Smirnova (2006: 301)]

- (9) ***Wenn** er aber finden **wird**, daß Jhr Euer Hertz nicht zu ihm*
when he but find.INF werden.3.SG that you your heart not to him
*lencken wollet, **nachdem** Jhr von allem fogenannten Zwange*
lead.INF want.2.PL after you of all so-called constraint
*befreyet feyn **werdet**, so wird er aufhören, an*
free.PAST.PART be.INF werden.2.PL so werden.3.SG he stop about
Euch zu dencken.
you to.think.INF
‘But if he finds that you do not want to turn your heart to him, after you have been freed from all of the aforementioned constraints, he will stop thinking about you.’

[Richardson, 1748, DTA]

- (10) *fo will ich dasjenige, was der meinigen fehlt, sobald es*
 so want.1.SG I that what from mine miss.3.SG as soon as it
möglich feyn wird, hinzufügen
 possible be.INF werden.3.SG add.INF

‘I will add what is missing from mine as soon as it will be possible.’

[Gessner, 1745, DTA]

It should be noted that the temporal subordinators *nachdem* ‘after’ and *sobald* ‘as soon as’ are much younger than the temporal-conditional *wenn* ‘when’: *Nachdem* is first attested since ca. 1550, and *sobald* appears at around 1600, whereas *wenn* ‘when’ was present since the Old High German period. In the context of this study, only the temporal and realis-conditional uses of the temporal-conditional subordinator *wenn* are relevant, as only those are associated with the development of the construction *werden* + infinitive.⁴ These subordinators share a general conceptual schema that is relevant for the history of *werden* + infinitive that is investigated in this study.

Essentially, the general temporal schema of these contexts may be conceptualized as follows. The scene consists of two events: the event *e1* that takes place at time *t1* and the event *e2* at time *t2*. The event *e1* is expressed in the subordinate clause with the subordinators *wenn* ‘when’, *nachdem* ‘after’, and *sobald* ‘as soon as’, and the event *e2* is depicted in the main clause that usually follows the subordinate clause. The time *t1* precedes the time *t2*, what is often iconically represented in the syntagmatic ordering of the clauses; both events are usually located in the future relative to the speech time.

Importantly, in contexts like (8)–(10), *werden* + infinitive can only be read as future tense construction. That is, it cannot be read in one of its present readings mentioned in Section 1, i.e. neither as a modal nor as an evidential verb, as these markers are not compatible with subordinate clauses.

The expansion of *werden* + infinitive into temporal and conditional subordinate clauses along the trajectory presented above thus represents a very advanced step on a grammaticalization path towards future tense marker. From the usual way grammaticalization processes proceed, it is to be expected that in the further stages of this process, the construction *werden* + infinitive should get firmly established in these contexts and become obligatory.

⁴Whereas irrealis-conditional contexts with *wenn* are clearly marked by subjunctive verbal morphology, realis-conditional contexts with indicative verbal form are often ambiguous between temporal and conditional reading throughout the time period.

However, from what we know about present-day German, *werden* + infinitive is no longer used in these contexts.⁵ The following case study is based on data consisting of conditional and temporal clauses, in order to trace recent changes in the development of the construction *werden* + infinitive and see what happened in the time between ENHG and today.

3.1 Data

The present study is based on data from three recent diachronic slices of German: 1600–1649, 1700–1749, and 1950–1999. The data comes from two corpora of German: Deutsches Textarchiv (DTA) for the first half of the 17th and of the 18th centuries, and the DWDS-Kernkorpus for the second half of the 20th century, which in my data represents present-day use. Table 1 gives an overview of the total occurrences of temporal and conditional clauses in the data. The number of subordinators *wenn* ‘when/if’, *nachdem* ‘after’, and *sobald* ‘as soon as’, tagged in the corpus as KOUS ‘subordinating conjunction’, is taken as a proxy for the number of the clauses they introduce.

Table 1: Subordinate clauses with *wenn*, *nachdem*, and *sobald*

period	corpus	corpus size	<i>wenn</i> clauses	<i>nachdem</i> clauses	<i>sobald</i> clauses
1600–1649	DTA	7 203 918	9 462	347	5
1700–1749	DTA	23 500 609	64 483	7 767	437
1950–1999	DWDS	58 068 993	88 782	5 548	2 147

Figure 1 depicts the normalized frequencies per million words of the same types of subordinate clauses. As can be seen, *wenn* is the most frequent subordinator across the three periods, the other two being far less frequent.

The qualitative analysis is mainly based on the data presented in Table 2, where only subordinate clauses containing the construction *werden* + infinitive are shown. The data amounts to a total of 740 observations that are distributed over three periods of time and the three subordinators under investigation.

⁵The fact that *werden* + infinitive occurred in these contexts at all is particularly interesting from the cross-linguistic point of view. As is known, ‘real’ future tense constructions behave differently with respect to this type of subordinate clause. The English *will*-future or the synthetic future in French are banned from conditional clauses, whereas in Slavic languages, future markers appear naturally in these contexts.

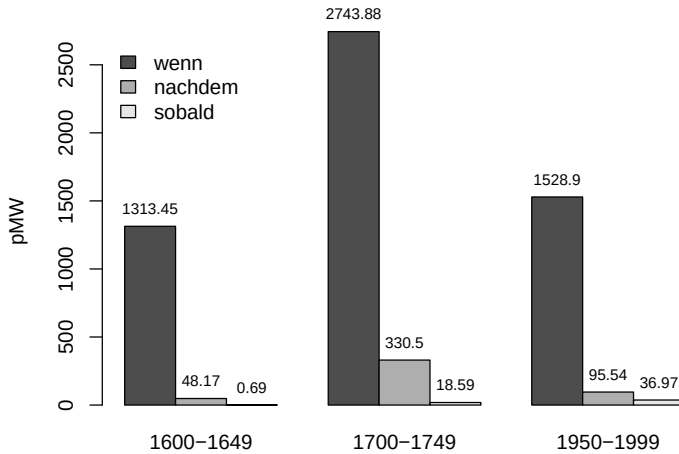


Figure 1: *wenn*, *nachdem*, and *sobald*, normalized frequency

Table 2: Subordinate clauses with *werden* + infinitive

period	corpus	<i>wenn</i> clauses	<i>nachdem</i> clauses	<i>sobald</i> clauses
1600–1649	DTA	213	2	0
1700–1749	DTA	312	23	4
1950–1999	DWDS	162	10	14
		687	35	18

3.2 Analysis

Figure 2 shows the relative frequency of the clauses containing the construction *werden* + infinitive in the three types of subordinate clauses.

The numbers in Figure 2 reveal that the proportions are extremely low: Most bars hardly reach the one percent mark. That means that subordinate clauses with *werden* + infinitive are extremely rare. In the first period, *wenn*-clauses have the highest proportion of *werden* + infinitive (2.25%). Interestingly, a decline in the use of *werden* + infinitive can be observed in all types of subordinate clauses; it is most pronounced in *wenn*-clauses (dark bars in Figure 2), but a similar progression can be observed for *nachdem*- and *sobald*-clauses. What we see here is a clear tendency for *werden* + infinitive to slowly disappear from the conditional

4 German *werden*-future: A case of failed grammaticalization?

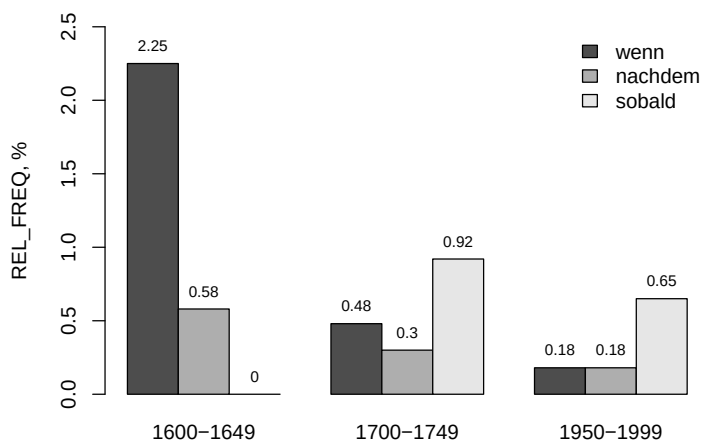


Figure 2: Relative frequency of *werden* + infinitive in subordinate clauses with *wenn*, *nachdem*, and *sobald*, in percent

and/or temporal clauses in the course of time. If we look at the younger subordinators *nachdem* and *sobald*, we can see that *werden* + infinitive initially finds its way into these structures, but fails to gain ground and declines in use in the subsequent periods.

From this, it can be concluded that *werden* + infinitive gradually loses its contexts of use. In Section 3, conditional and temporal subordinate clauses were described as representing an advanced stage of grammaticalization of *werden* + infinitive as a future tense marker. These contexts were the last into which the construction expanded, and the further grammaticalization process would normally imply further expansion and consolidation of use in these contexts. What we observe, however, is the opposite – *werden* + infinitive is gradually withdrawing from these contexts. The construction does not proceed further on the grammaticalization path.

Furthermore, a closer look into the data reveals that, in the third period, the subordinate clauses differ considerably from each other as concerns the form of the infinitive (see Table 3).

Whereas *wenn*-clauses still predominantly contain *werden* + infinitive (see e.g. 11), *nachdem*- and *sobald*-clauses show a clear preference for *werden* + infinitive perfect, usually called future perfect or *Futur II* in German (see e.g. 12).

Table 3: Form of the infinitive, 1950–1999

subordinator	<i>wenn</i>	<i>nachdem</i>	<i>sobald</i>
<i>werden</i> + infinitive	137	4	1
<i>werden</i> + infinitive perfect	25	10	9
% infinitive perfect	15%	71%	90%

- (11) *Wenn wir trotzdem an diesen ungesicherten Allgemeinbegriffen*
when we nonetheless to these unsecured generalities
festhalten werden, so liegt darin mehr als ein unentschuldbarer
cling.INF werden.3.PL so lie.3.SG there more than a inexcusable
methodologischer Leichtsinns.
methodological recklessness
‘If we will nevertheless cling to these unsecured generalities, there is
more than an inexcusable methodological recklessness in this.’
[Dahrendorf, 1957, DWDS-Kernkorpus]
- (12) *Sie hat versprochen sich zu entscheiden, sobald sie das Leichentuch*
she has promised REFL to decide as soon as she the shroud
für ihren Schwiegervater Laertes fertiggewebt haben wird
for her father-in-law Laertes finish.PAST.PART have.INF werden.3.SG
‘She has promised to decide as soon as she will have finished weaving the
shroud for her father-in-law Laertes.’
[Schwanitz, 1999, DWDS-Kernkorpus]

The form *werden* + perfect infinitive is usually employed to mark a completion of an event in the future and is particularly tied to contexts of temporal clauses such as those analyzed here.

Another important observation concerns the verb form in the main clauses. The relative proportion of different verb forms is depicted in Figure 3.

What we see in Figure 3 is a clear tendency towards the decline of the form *werden* + infinitive across the three periods: Whereas in the 17th century, *werden* + infinitive was the dominant form in both clauses of the sentence, in the 20th century, *werden* + infinitive in the main clause amounts only to 40% of all verb forms. On the other hand, the rise of the use of simple present forms is particularly pronounced: The use of the present tense increases from ca. 10% in the first period to 40% in the third period. That is, main clauses follow the same general

4 German *werden*-future: A case of failed grammaticalization?

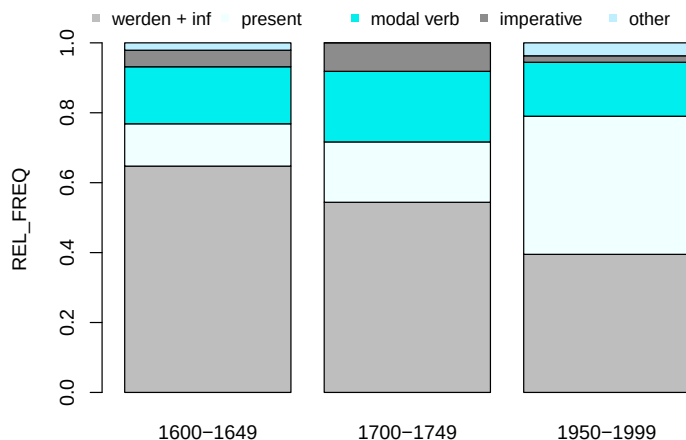


Figure 3: Relative frequency of different verb forms in main clauses, in percent

tendency we have observed above, and the construction *werden* + infinitive gets gradually replaced by the present tense forms, both in main and in subordinate clauses of temporal and conditional sentences.

The data presented so far shows that *werden* + infinitive gradually disappears from temporal and conditional clauses with *wenn*, *nachdem*, and *sobald* (see Figure 2). In present-day German, *werden* + infinitive is almost completely absent from the temporal clauses with *nachdem* and *sobald* (see Table 3); *werden* + perfect infinitive (= “future perfect”) is used there instead. As seen from Figure 2, *werden* appears first in these contexts, but fails to gain ground and disappears gradually from them. Moreover, this downward tendency holds for main clauses of these sentences as well (see Figure 3).

A last observation to be drawn from the analysis of this dataset concerns the infinitive verbs combining with *werden*.⁶ Table 4 gives an overview of the ten most frequent infinitive verbs in each period.

As can be seen from Table 4, some of the infinitive verbs, such as *kommen* ‘come’ or *haben* ‘have’, have combined with *werden* from the beginning; others have become more or less associated with the construction in the course of time. In the verb list of the last period, however, *sein* ‘be’ is not only the most frequent verb, but it is also almost the only verb in this combination. With 64 occurrences,

⁶For the infinitive verbs, only the forms with the present infinitive, and not those with the perfect infinitive, were analyzed.

Table 4: Infinitive verbs, top ten collocates

1600–1649			1700–1749			1950–1999		
verb	gloss	N	verb	gloss	N	verb	gloss	N
<i>kommen</i>	come	33	<i>kommen</i>	come	41	<i>sein</i>	be	64
<i>erscheinen</i>	appear	18	<i>sein</i>	be	28	<i>haben</i>	have	9
<i>sein</i>	be	16	<i>haben</i>	have	19	<i>kommen</i>	come	4
<i>anschauen</i>	look	5	<i>erscheinen</i>	appear	13	<i>gelingen</i>	succeed	3
<i>offenbaren</i>	reveal	5	<i>sehen</i>	see	9	<i>lesen</i>	read	3
<i>sagen</i>	say	5	<i>werden</i>	become	8	<i>werden</i>	become	3
<i>sehen</i>	see	5	<i>hören</i>	hear	7	<i>ausmachen</i>	turn out	2
<i>werden</i>	become	5	<i>posaunen</i>	broadcast	7	<i>führen</i>	lead	2
<i>angehen</i>	concern	4	<i>aufhören</i>	stop	5	<i>gehen</i>	go	2
<i>brechen</i>	break	4	<i>offenbaren</i>	reveal	5	<i>heiraten</i>	marry	2

it stands in sharp contrast to other infinitive verbs, which are attested in the data with very rare occurrences.

Hartmann (2021: 385) analyzed the usage of the construction *werden* + infinitive in the ENHG period and found that towards the end of the period, i.e. in the 17th century, the most frequent infinitive verb in this construction is *sein* (23 occurrences in his data), followed by *haben* ‘have’ (with 8 occurrences). In a study based on data from the 19th and 20th century, Hilpert (2008: 148) detected the same verbs, that is, *sein* ‘be’ (n=172), *kommen* ‘come’ (n=62), *haben* ‘have’ (n=40), and *werden* ‘become’ (n=40) among the most frequent infinitive verbs in this construction. In each study, it is thus the verb *sein* that turns out to be the most frequent infinitive verb that combines with *werden*. This suggests that the changes detected above and that concern the temporal and conditional clauses are not only characteristic of these particular contexts, but manifest a general tendency in the development of the construction *werden* + infinitive.

In sum, the results of the diachronic corpus study show that in the last four centuries, the construction *werden* + infinitive has not progressed further on its way towards becoming a grammaticalized future tense marker:

1. With regard to usage frequency, there is a decrease in use and a gradual replacement of *werden* + infinitive by the futurate present tense. As concerns its use in temporal and conditional subordinate clauses, after initial uses, it failed to expand further, and this tendency holds also for independent uses of this construction. That is, no further context-expansion of the construction in the sense of Himmelman (2004) can be observed since ENHG. Instead, the contexts of use become more and more restricted.

2. With regard to the infinitive verbs used in the construction, the range of regularly used verb types decreased, most of them being used rarely in the construction. Importantly, *werden* + infinitive established a particular preference for the infinitive verb *sein* 'be'. This limitation to one dominant usage pattern with *sein* clearly goes against the tendency of obligatorification (Lehmann 1995). During the advanced stages of a grammaticalization process, obligatorification would manifest itself in the increasing token and type frequencies of the infinitive verbs. However, this is clearly not the case for *werden* + infinitive, as shown above.

To conclude this section, in the light of the evidence presented above, I would like to argue that the development of the construction *werden* + infinitive in the last four centuries constitutes a case of failed grammaticalization. What has happened instead is that *werden* + infinitive has limited itself to some non-frequent and specialized uses. This empirical evidence points to the establishment of a possible functional niche of this construction in modern German. In Section 4, I will deal in more detail with the combination of *werden* + *sein* to see what kind of functional niche this particular pattern occupies in present-day German.

4 The functional niche of *werden* + infinitive in present-day German

Let me start this section by giving some illustrative examples of the combination of *werden* and *sein* (cf. 13–15).

- (13) *Du wirst mir eines Tages dankbar sein.*
 you werden.2.SG me one day thankful be.INF
 'You will be grateful to me one day.'

[Noll, 1996]

- (14) *Was wird inzwischen zu Hause alles geschehen sein?*
 what werden.3.SG meanwhile at home all happen.PAST.PTCPPL sein.INF
 'What will have happened at home in the meantime?'

[Grzimek, 1954]

- (15) *Ob und wie dieser Ratschlag funktionierte, darüber wird noch zu sprechen sein.*
 whether and how that advice work.3.SG.PAST about it werden.3.SG
 yet to speak.INF be.INF
 'Whether and how this advice worked, will have to be discussed yet.'

[P. M., 1993]

The examples (13)–(15) show that the combination of *werden* + *sein* occurs in different syntagmatic configurations and displays an array of possible functions. In (13), *sein* is a copula verb that combines with an adjective (*dankbar* ‘grateful’); the combination with *werden* renders future time reference. In (14), on the other hand, *sein* is a perfect tense auxiliary modifying the full verb *geschehen* ‘happen’, and the combination *werden* + *past participle* + *inf* constitutes a case of Future II or future perfect. This combination may refer to future time, but mostly, it is interpreted as a modal construction, i.e. expressing the epistemic uncertainty of the speaker. In (15), *sein* appears as a part of another construction, i.e. *sein* + *zu* ‘to’-infinitive with the modal meaning of obligation. The combination with *werden* can be interpreted as referring to an obligation that will hold at a later point in time, or as a textual lookahead, where the author announces that something will (have to) be treated later in the text.

Given this variety of functions the syntagmatic combination of *werden* and *sein* may carry in individual contexts, it is worthwhile to inspect the individual contexts in more detail in order to see whether there are patterns in the data that point to the existence of a particular functional niche of the pattern *werden* + *sein*. This is the aim of the following synchronic case study.

4.1 Data

The present analysis is based on a random sample of 1000 observations that stem from the most recent slice of the DWDS-Kernkorpus20 (Geyken 2007), i.e. from the years 1950–1999.⁷ The data were manually annotated with respect to the type of the element the infinitive verb *sein* combines with. The following categories were used for annotation:

- ADJECTIVE (see 13 above)
- NOUN PHRASE (see 16)
- PREPOSITIONAL PHRASE (see 17)
- ADVERB (see 18)
- ZU-INFINITIVE (see 15 above)
- PAST PARTICIPLE (see 14 above)
- (ADJECTIVAL) PAST PARTICIPLE (see 19)

⁷The search string used to extract the examples of the combinations of *werden* and *sein* from the corpus was the following: /werde|wirst|wird|werden|werdet/ && “sein with \$p=/V.*INF/” #asc_date[1950-00-00, 1999-99-99]; i.e. I searched for all present indicative forms of the verb *werden* in combination with the infinitive form of *sein* within a sentence. The obtained hits were then randomized, manually inspected for false positives and annotated according to the variables described in this section.

- (16) *Das ausführende Organ der Bank wird der Vorstand*
the executive body of the bank werden.3.SG the board of directors
sein.
be.INF
‘The executive body of the bank will be the board of directors.’
[AdG 40, 1970]
- (17) *Ich bin zu Hause, wie gesagt, und solange es hell ist,*
I be.1.SG at home as said and as long as it light be.3.SG
werde ich im Garten sein.
werden.1.SG I in the garden be.INF
‘I am at home, as I said, and as long as it is light outside, I will be in the garden.’
[Born, 1979]
- (18) *Peter wird natürlich auch da sein.*
Peter werden.3.SG of course too there be.INF
‘Peter will of course be there too.’
[Horster, 1950]
- (19) *Dann, und nur dann, wird ein erster Schritt auf dem langen Wege*
then and only then werden.3.SG a first step on the long way
zur nuklearen Abrüstung getan sein.
to nuclear disarmament do.PAST.PTCPPL be.INF
‘Then, and only then, will a first step on the long road to nuclear disarmament be taken.’
[AdG 28, 1958]

In combinations with adjectives (13), noun phrases (16), prepositional phrases (17), and adverbs (18), *sein* qualifies as a copula verb. For this reason, these combinations will be treated further below as one sub-group of *sein*-usages. As noted above, the combination of *sein* with *zu* ‘to’-infinitive qualifies as a construction on its own, and therefore it will be treated as such in the following. In the contexts like (14), *sein* will be considered a perfect auxiliary.

In combinations like (19), *sein* is usually classified as passive auxiliary, and the construction *sein* + past participle is called stative passive or *Zustandspassiv* (cf. e.g. Helbig 1983; Duden 2016: 567–568). However, the grammatical classification of the construction as stative passive is not uncontroversial. Without going into much detail at this point, I will follow here the account of Maienborn (2007) who

has convincingly shown that the past participle in this construction behaves like an ordinary adjective, and *sein* in these combinations can be considered a “normal” copula verb. For this reason, I will treat such combinations as pertaining to the sub-category of copula verb *sein* mentioned above. Table 5 gives the frequency of these elements in the sample.

Table 5: Complements of *sein*-infinitive

complement	N
adjective	388
noun phrase	264
<i>zu</i> -infinitive	85
adjectival participle	81
prepositional phrase	81
adverb	54
past participle	47
TOTAL	1000

Figure 4 summarizes the combinations of *werden* and the infinitive of *sein* according to the three categories of the verb *sein* introduced above:

- copula verb in combination with adjectives, adjectival past participles, noun phrases, prepositional phrases, and adverbs
- modal construction *sein* + *zu* ‘to’-infinitive and
- perfect auxiliary in combination with past participles of full verbs.

As can be already seen from Table 5 and Figure 4, the distribution of different variants of the infinitive verb *sein* is extremely skewed.⁸ In the next section, I will look more closely into the functional range of these combinations.

4.2 Analysis

In the preceding section, three categories of the verb *sein* were distinguished according to the elements the verb combines with, and to the semantics the whole syntagmatic configuration with *werden* conveys. Let me start by discussing the two categories that are only sparsely presented in the data and make up 13.2% of the analyzed sample.

⁸Even if I had considered the 81 cases of so-called *Zustandspassiv* as a separate category, this would not have considerably changed the distribution.

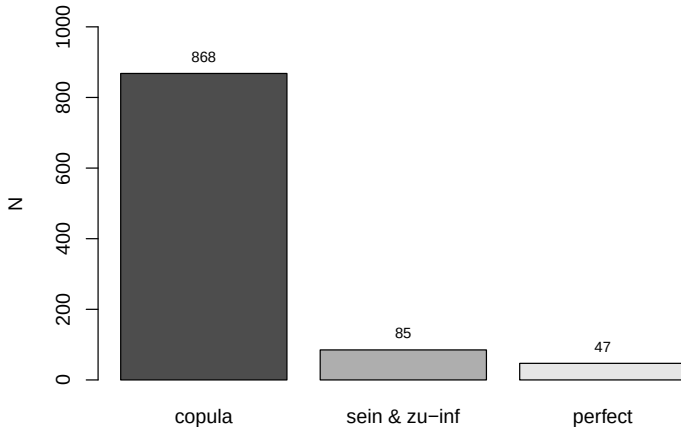


Figure 4: Frequency of different *sein*-variants in the data

4.2.1 The future perfect form *werden* + past participle + *sein*

The future perfect form is attested 47 times in the sample. As mentioned in Table 5 and Figure 4, in this context *werden* functions as a modal-evidential verb, and it cannot be considered exclusively a future tense auxiliary. In this reading, *werden* behaves similarly to other epistemically used modal verbs like *müssen* ‘must’ or *dürfte* ‘may’ (see 20a–20b and 21a–21b):

- (20) a. *Es **wird** wohl ’ne größere Taube gewesen sein.*
 it werden.3.SG probably a larger pigeon be.PAST.PTCPL be.INF
 ‘It will probably have been a larger pigeon.’
 [Bieler, 1983]
- b. *Es **muss/dürfte** wohl ’ne größere Taube*
 it must.3.SG/may.3.SG.SUBJ probably a larger pigeon
gewesen sein
 be.PAST.PTCPL be.INF
 ‘It must have been a larger pigeon.’
- (21) a. *Für die “Verräter” **wird** dieser Rausschmiß nicht unerwartet*
 for the traitors werden.3.SG this expulsion not unexpectedly
gekommen sein.
 come.PAST.PTCPL be.INF
 ‘For the “traitors” this expulsion will not have come unexpectedly.’
 [ZEIT, 1954]

- b. Für die “Verräter” **muss/dürfte** dieser Rausschmiß nicht unerwartet
 for the traitors werden.3.SG this expulsion not unexpectedly
 gekommen sein
 come.PAST.PTCPL be.INF
 ‘For the “traitors” this expulsion should not have come unexpectedly.’

These combinations are very low in frequency (see Figure 4 in Section 4.1). Moreover, they are functionally similar to other means of expressing epistemic and/or evidential assessment in present-day German. Furthermore, future time reference, i.e. future in the past, constitutes only one possible interpretation of these combinations, whereas the modal nuances are much more pronounced.

This suggests that this usage of *werden* + *sein* does not represent a special functional niche of the more abstract construction *werden* + infinitive. Rather, the future perfect forms attested in the analyzed sample should be seen in a broader context of the usage of perfect infinitive forms in present-day German, including perfect infinitive forms with the perfect auxiliary *haben*, which in many contexts of use display very similar formal and functional characteristics.

4.2.2 *Werden* and the modal construction *sein* + *zu* ‘to’-infinitive

The combination of *werden* with the modal construction *sein* + *zu* ‘to’-infinitive amounts to 85 observations in the analyzed sample (= 8.5%). What makes these contexts particularly interesting is the fact that the combination with *werden* unambiguously renders a future tense reading. Compare for example (22a) and (23a) with (22b) and (23b) respectively, where *werden* has been removed, and only the present tense form of *sein* + *zu* ‘to’-infinitive is present.

- (22) a. Es **wird** nie wieder gutzumachen sein.
 it werden.3.SG never ever to.make.good be.INF
 ‘It will never be possible to make up for it.’

[Bachmann, 1971]

- b. Es **ist** nie wieder gutzumachen.
 it be.3.SG never ever to.make.good.INF
 ‘It can never be made up for.’

- (23) a. *Ihre Aufenthaltserlaubnis wird jedes Jahr zu erneuern*
 their residence permit werden.3.SG every year to renew.INF
sein, und ihre Arbeitserlaubnis wird jedes Mal zu
 be.INF, and their work permit werden.3.SG every time to
erneuern sein, wenn ...
 renew.INF be.INF when ...
 ‘Their residence permit will have to be renewed every year, and their
 work permit will have to be renewed every time when ...’
 [AdG 39, 1969]
- b. *Ihre Aufenthaltserlaubnis ist jedes Jahr zu erneuern, und ihre*
 their residence permit be.3.SG every year to renew.INF, and their
Arbeitserlaubnis ist jedes Mal zu erneuern, wenn ...
 work permit be.3.SG every time to renew.INF when ...
 ‘Their residence permit must be renewed every year, and their work
 permit must be renewed every time when ...’

In (22b) and (23b), the future tense reading gets lost, and only the modal semantics of the construction *sein* + *zu* ‘to’-infinitive remains, which in turn may be read as a general, tenseless predication. In opposition to (22a) and (23a), however, it is clear that the added value of *werden* consists of conveying future tense semantics. That is, the modal semantics of the construction *sein* + *zu* ‘to’-infinitive is compositionally combined with the future tense semantics of *werden*. Thus, in these contexts, we find *werden* in clear opposition to the simple present tense, and in these contexts, the simple present is not able to produce clear futurate readings.

To sum up so far, the combination of *werden* and a conventionalized construction *sein* + *zu* ‘to’-infinitive with its own modal passive meaning serves to unambiguously mark the future time reference of the proposition. In these contexts, *werden* + infinitive is not functionally equivalent to the present tense, as the present tense form of the construction does not allow for a future tense interpretation. I will return to this point in Section 4.3.

4.2.3 *Werden* and the copula verb *sein*

The great majority of observations of *werden* + *sein* contains the infinitive of the copula verb *sein*, i.e. 868 observations (= 86.8%). Only in a small fraction of these observations, namely in 12 sentences, does *werden* not primarily evoke a future tense interpretation, but may be attributed a modal-evidential reading (see 24–25). Note that in these contexts, modal adverbials such as *wohl* ‘surely/probably’

are present in the text and contribute as well to the modal meaning of the proposition.

- (24) *Was bei Mikrowellen, Joysticks und Zahnbürsten erlaubt ist, **wird** doch wohl bei der Soziologie nicht verboten **sein**.*
what for microwaves joysticks and toothbrushes allow.PAST.PTCPL
be.3.SG werden.3.SG MP⁹ surely for the sociology not forbidden be.INF
'What is allowed for microwaves, joysticks and toothbrushes will surely not be forbidden for sociology.'
[ZEIT, 1996]

- (25) *Das **wird** doch nicht etwa schon die Polizei **sein**?*
that werden.3.SG MP not MP already the police be.INF
'Surely that won't be the police already?'
[Grass, 1959]

In the vast majority of cases, however, the future temporal interpretation is clearly the dominant reading present in the context. Sometimes (in 196 out of 856 or in ca. 23%), temporal adverbials or other explicit contextual information with future time interpretation are present, as in (17)–(19) and (26), but this certainly does not need to be the case (see 16 and 27).

- (26) *Im Jahr 2005 **wird** Sprache im Umgang mit dem Computer in in year 2005 werden.3.SG language in handling with the computer in jeder Hinsicht verfügbar **sein**.*
every aspect available be.INF
'In 2005, language will be available in every aspect of computer use.'
[BILD, 1998]

- (27) *Viele potentielle Produzenten **werden** nicht in der Lage **sein**, das many potential producers werden.3.PL not able be.INF the erforderliche Kapital zu erhalten.*
necessary capital to obtain.INF
'Many potential producers will not be able to obtain the necessary capital.'
[AdG 46, 1976]

⁹German modal particles (MP) are notoriously difficult to translate. For this reason, I resort to the label MP in the glosses.

It may be concluded that for the combinations of *werden* with the copula verb *sein*, the function of *werden* unambiguously and almost exclusively consists of marking future time reference. It is in these contexts that the future meaning of the construction *werden* + infinitive is highly distinctive from the other contexts of use.

The frequency-based evidence presented so far is already very telling: (i) *Sein* is by far the most frequent infinitive verb that combines with *werden* in present-day German (see Section 3.2); (ii) it is the copula verb *sein* that mostly combines with *werden* (see Section 4.1). This suggests that the combination of *werden* and the copula verb *sein* constitutes a very significant use of *werden* + infinitive and may be qualified as a niche. The qualitative evidence presented in this section certifies that in these contexts, *werden* functions as a future tense marker. In other words, the future tense reading of *werden* + infinitive is largely restricted to the combinations of *werden* and *sein*, whereas in all other contexts, it is the (futurate) present tense that is employed to mark future tense reference. In Section 4.3, I will present a tentative account of why combinations with the copula *sein* represent a suitable context for future uses of *werden* + infinitive in present-day German.

4.3 Discussion

It is generally acknowledged that German has three copula verbs: *sein*, *werden*, and *bleiben*. In present-day German, the copula verb *sein* is a stative copula used to ascribe qualities to all kinds of referents, and it stands in semantic opposition to two further copula verbs, i.e. to the inchoative *werden* on the one hand, and to the durative stative *bleiben* on the other, as in (28).

- (28) *Lena ist/wird/bleibt* *böse_{Adj}/meine Freundin_{NP}*.
 Lena be./werden./remain.3.SG angry/my friend
 ‘Lena is/becomes/remains angry/my friend.’

In the following, only the pair *sein* vs. *werden* will be relevant for our purposes. The semantic difference between the two is one of aktionsart: Whereas *sein* is stative and atelic, *werden* is inchoative and telic. These copula verbs are usually seen to behave very similarly with respect to the predicative elements they combine with – that is, they may combine with adjectives, with noun phrases, with prepositional phrases, and with adverbs (see Section 4.1).

However, there is an important difference in syntactic behaviour between the two. The range of the usage contexts of the copula *werden* is more restrictive than those of *sein*. Whereas *sein* may be combined with all types of predicative

elements, irrespective of their semantics, the use of *werden* seems to be restricted to complements that express incremental changes. That is, the use of *werden* is ruled out in combination with locative expressions such as *im Garten* ‘in the garden’ or with non-gradable adjectives such as *tot* ‘dead’. Table 6 summarizes these differences.

Table 6: Complements of *sein*- vs. *werden*-copula

complement	<i>sein</i>	<i>werden</i>	gloss
gradable property	A ist böse.	A wird böse.	‘angry’
non-gradable property	A ist tot.	*A wird tot.	‘dead’
location	A ist im Garten.	*A wird im Garten.	‘in the garden’
location	A ist dort.	*A wird dort.	‘there’

These semantic differences manifest themselves in an interesting way when it comes to future time reference. Combinations with the copula *sein* unambiguously render stative interpretations, and usually locate the ascribed state or property in the present time. This applies to all kinds of expressions, irrespective of their gradable/non-gradable nature, as in (29)–(30).

- (29) *Er ist (immer) böse, wenn er das hört.*
 he be.3.SG (always) angry when he it hear.3.SG
 ‘He is always angry when he hears that.’

- (30) *Sie ist Astronautin.*
 she be.3.SG astronaut
 ‘She is an astronaut.’

Combinations with the copula *werden* with complements that refer to gradable properties or imply incremental changes, on the other hand, are essentially ambiguous between the inchoative reading with present time reference (= ongoing change of state) and the future reading (= result state in the future). It is only by means of further contextual elements such as temporal adverbials that a non-ambiguous present or future reading is achieved, as in (31a–31b).

- (31) a. *Er wird langsam böse.*
 he werden.3.SG slowly angry
 ‘He is getting angry.’

- b. *Er wird böse, wenn er das hört.*
 he werden.3.SG angry when he it hear.3.SG
 'He will be/get angry when he hears that.'

Combinations of the auxiliary *werden* with the copula *sein*, on the other hand, clearly result in a non-ambiguous future tense reading, as in (32):

- (32) *Er wird böse sein.*
 he werden.3.SG angry be.INF
 'He will be angry.'

For the expressions that refer to non-gradable properties or non-incremental changes, the situation is rather different. They may be combined with the copula *sein* to refer to a present state,¹⁰ as in (33). When combined with the auxiliary *werden* and the infinitive of the copula *sein*, they unambiguously refer to a future state, as in (34). Note that the option with the copula verb *werden* alone is not available (see Table 6).

- (33) a. *Sie ist bereit.*
 she be.3.SG ready
 'She is ready.'
- b. *Sie wird bereit sein.*
 she werden.3.SG ready be.INF
 'She will be ready.'
- (34) a. *Ich bin im Garten.*
 I be.1.SG in.the garden
 'I am in the garden.'
- b. *Ich werde im Garten sein.*
 I werden.1.SG in.the garden be.INF
 'I will be in the garden.'

From what has been observed in previous sections, and from the semantic analysis just presented, it is reasonable to assume that for present-day German copula verbs, there exists a firmly established paradigmatic pair *sein* with present

¹⁰The copula verb *sein* in its present tense form is in principle suited for futurate contexts and can be used to refer to future states, especially if some adverbial modification is present in the text that unambiguously refers to future, cf. *Morgen ab 16 Uhr bin ich im Garten* 'Tomorrow from 4 p.m. I will be in the garden.' However, temporal adverbials are obligatory in such contexts.

time reference and *werden* + *sein* with future time reference. Though *werden* is essentially an inchoative copula verb and can be used as such in contexts with gradable expressions, its use in the construction *werden* + *sein* has been conventionalized as strong expression of future tense, in opposition to the present tense of the copula *sein*.

This paradigmatic relation is strongly pronounced in case of more or less fixed multi-word expressions containing the copula verb *sein*. These include, for example, conventionalized and idiomatic expressions such as *an der Reihe sein* ‘be next, lit: be in the turn’, *in der Lage sein* ‘be able to, lit: be in the position’, *zu Ende sein* ‘be finished’, and the like. The only way to mark future time reference is to employ the future auxiliary *werden*, as in (35)–(36).

- (35) *Unsere Kultur* **wird** *in der Lage sein, sich unabhängig von*
our culture werden.3.SG able be.INF REFL independently of
der menschlichen Biologie und ihren Gesetzen zu entwickeln.
the human biology and its laws to develop.INF
‘Our culture will be able to develop independently of human biology and its laws.’

[Rechenberg, Informatik, 1991]

- (36) *Und auch hier* **wird** *es von entscheidender Bedeutung sein, wie*
and too here werden.3.SG it of critical significance be.INF how
sich die DDR-Führung verhält.
REFL the GDR leadership behave.3.SG
‘And here, too, it will be crucial how the GDR leadership behaves.’

[DEDok., 1990]

To sum up, I would like to argue that the future construction *werden* + infinitive have “survived” in this particular niche – that is, in combination with the infinitive of copula verb *sein*. In combination with other infinitive verbs, *werden* + infinitive did not develop further, and the futurate present tense was more successful in expressing future time reference.

In this diachronic situation, the contexts with the copula verb *sein* offered a welcome niche for the future construction *werden* + infinitive. And this is, as I believe, at least partly due to the asymmetries of usage contexts of the copula verb *werden* described above. As discussed above, *werden* as a copula cannot be used in present tense contexts with a significant number of complements due to their non-incremental and non-gradable semantics. The future auxiliary in combination with the copula *sein* is, however, fine. It could be hypothesized that

the combination of *werden* and *sein* serves as a sort of compensating strategy for this gap. Once established in this particular context, it could survive there and be preserved till the present-day German. Of course, more research is needed to substantiate this idea.

5 Conclusions

In this paper, I have argued that the development of the future construction *werden* + infinitive constitutes a case of failed grammaticalization. Although the relatively high grammaticalization status of the construction seems uncontested in present-day German, especially when semantic and formal characteristics of this construction are concerned, a closer look at the frequency of use and the distribution over different infinitive verbs offers a more differentiated picture. Namely, these factors suggest that the grammaticalization process has not come to its full completion, leaving *werden* + infinitive “just” an alternative means of expressing futurity.

Not being successful and leaving the grammaticalization path does not need to mean, however, that the construction dies out and disappears. As this study has shown, *werden* + infinitive has survived in a particular context of use, i.e. in combination with the infinitive of the copula verb *sein*. In these contexts, the auxiliary *werden* has become nearly obligatory, and it serves here to mark future time reference. The synchronic motivations of this situation as well as the diachronic details of this process are complex and concern not only the construction *werden* + infinitive itself but also the related copula verb *werden*. In Section 4.3, I offered a first tentative account of this functional niche of *werden* in present-day German. But of course, this idea needs to be followed up in further research and substantiated with more data, synchronically and diachronically.

References

- Brons-Albert, Ruth. 1982. *Die Bezeichnung von Zukünftigem in der gesprochenen Deutschen Standardsprache*. Tübingen: Narr.
- Diewald, Gabriele & Mechthild Habermann. 2005. Die Entwicklung von *werden* + Infinitiv als Futurgrammem. In Torsten Leuschner, Tanja Mortelmans & Sarah Groodt (eds.), *Grammatikalisierung im Deutschen* (Linguistik - Impulse und Tendenzen 9), 229–250. Berlin: de Gruyter.

- Diewald, Gabriele & Elena Smirnova. 2010. *Evidentiality in German. Linguistic realization and regularities in grammaticalization* (Trends in Linguistics 228). Berlin: de Gruyter.
- Duden. 2016. *Die Grammatik*, vol. 4. Mannheim: Dudenverlag.
- Geyken, Alexander. 2007. The DWDS corpus: A reference corpus for the German language of the 20th century. In Christiane Fellbaum (ed.), *Collocations and idioms: Linguistic, lexicographic, and computational aspects*. London: Continuum. <https://www.dwds.de/d/korpora/kern>.
- Hartmann, Stefan. 2021. Diachronie der Zukunft: werden + Infinitiv und Konkurrenzkonstruktionen im Mittelhochdeutschen und Frühneuhochdeutschen. *Beiträge zur Geschichte der deutschen Sprache und Literatur* 143. 364–395.
- Helbig, Gerhard. 1983. Zustandspassiv, sein-Passiv oder Stativ? In Gerhard Helbig (ed.), *Studien zur deutschen Syntax*, vol. 1, 47–57. Leipzig: VEB (Verlag Enzyklopädie).
- Hilpert, Martin. 2008. *Germanic future constructions: A usage-based approach to language change*. Amsterdam: John Benjamins.
- Himmelman, Nikolaus. 2004. Lexicalisation and grammaticalization: Opposite or orthogonal? In Walter Bisang, Nikolaus Himmelmann & Björn Wiemer (eds.), *What makes grammaticalization? A look from its fringes and its components*, 21–44. Berlin: de Gruyter. DOI: 10.1515/97440.1.21.
- Ihden, Sarah. 2023. The Present Participle with *Wēren* and *Wēsen* in Middle Low German: A Corpus-Based analysis of structure and meaning. In Hanna Fischer, Melitta Gillmann & Mirjam Schmuck (eds.), *Encoding Aspectuality in Germanic Languages – Empirical and Theoretical Approaches (Journal of Germanic Linguistics)*, vol. 35/4, Special Issue, 409–446. Cambridge: Cambridge Univ. Press.
- Lehmann, Christian. 1995. *Thoughts on grammaticalization*. 1st edn. (LINCOM Studies in Theoretical Linguistics). München: LINCOM Europa.
- Maienborn, Claudia. 2007. Das Zustandspassiv. Grammatische Einordnung – Bildungsbeschränkung – Interpretationsspielraum. *Zeitschrift für germanistische Linguistik* 35(1-2). 83–114.
- Matzel, Klaus & Bjarne Ulvestad. 1982. Futur I und futurisches Präsens. *Sprachwissenschaft* 7. 282–328.
- Smirnova, Elena. 2006. *Die Entwicklung der Konstruktion würde + Infinitiv im Deutschen: Eine funktional-semantische Analyse unter besonderer Berücksichtigung sprachhistorischer Aspekte* (Studia Linguistica Germanica). Berlin: de Gruyter.
- Thieroff, Rolf. 1992. *Das finite Verb im Deutschen: Tempus – Modus – Distanz* (Studien zur deutschen Grammatik). Tübingen: Narr.

4 *German werden-future: A case of failed grammaticalization?*

Vater, Heinz. 1975. *Werden* als Modalverb. In Calbert, Joseph P. & Heinz Vater. *Aspekte der Modalität*. (Studien zur deutschen Grammatik 1). Tübingen: Narr.

